

wage bills, job creation or destruction, or composition of hires or separations, including early-career workers. These null results hold across occupations and persist even where encouragement is paired with enterprise chatbot solutions and training. While internal reorganization is underway, with workers reporting new AI-related tasks—especially in workplaces with chatbot initiatives—these changes have not translated into measurable differences in earnings or hours.

Overall, our findings challenge narratives of imminent labor market disruption from Generative AI. While we capture early impacts, two pieces of evidence suggest the limited labor market effects may persist in the foreseeable future. First, our dynamic difference-in-differences estimates remain flat throughout the two-year study period. Second, even among “AI front-runners”—workplaces that have adopted the full suite of proactive chatbot initiatives and workers who adopted early, use the tools daily, or report substantial benefits—the results remain null.

Contributions to the Literature. The key contribution of this paper is to link large-scale, representative data on AI chatbot adoption to labor market outcomes in administrative records. This linkage brings critical nuance to existing evidence on how Generative AI relates to labor market outcomes.

First, a contemporaneous set of studies uses exposure measures—technical assessments of which job tasks are feasible to assist with AI chatbots—to examine whether exposed occupations have fared differently after the release of ChatGPT. Brynjolfsson, Chandar and Chen (2025) document a sharp decline in employment of early-career workers in AI-exposed occupations in the United States since 2022, suggesting that Generative AI may already be disrupting labor markets. Our key contribution is to split these employment trends by whether firms have actually adopted Generative AI. While we replicate the U.S. pattern of declining early-career jobs in exposed occupations in Denmark, our difference-in-differences analysis reveals that the declines are not driven by firms adopting AI chatbots.

Second, a small but growing number of studies field worker-level surveys on the

adoption of Generative AI (Bick, Blandin and Deming, 2025; Hartley et al., 2025; Humlum and Vestergaard, 2025). While our adoption patterns align with these studies, our key contribution is to link survey responses to administrative data on labor market outcomes, thereby moving the analysis beyond self-reported effects.⁴

Third, while several experiments document sizable productivity gains from chatbot use on specific tasks (Brynjolfsson, Li and Raymond, 2025; Cui et al., 2024; Dillon et al., 2025; Noy and Zhang, 2023), it remains unclear how these gains translate into workers’ earnings and hours, as high-quality microdata on such outcomes are rarely available. While our occupation-wide surveys show more modest gains—about 3% time savings for the typical user—we still find null results even for workers and workplaces reporting substantial productivity improvements. Nonetheless, we find a clear association between adoption and occupational mobility, aligning with experimental evidence that chatbots provide greater benefits to workers with less prior expertise.

Fourth, we provide the first representative account of employer adoption of AI chatbots. Whereas surveys on traditional “pre-generative” AI show relatively low uptake (Acemoglu et al., 2022; Bonney et al., 2024), we find that most employers in exposed occupations have now embraced AI chatbots. Alternative firm-level data on Generative AI come from online job postings, where adoption is proxied by the listing of Generative AI skills in job descriptions (Lichtinger and Hosseini Maasoum, 2025; Schubert, 2025). While job postings indicate notable changes in such proxies after ChatGPT’s release, it is less clear how well they capture employees’ actual use of tools, especially non-specialized ones like chatbots. We address this gap by surveying workers on their chatbot use and employer initiatives, linking responses to administrative firm data. Although employer chatbot initiatives have not led to net changes in earnings or hours, they strongly predict the emergence of new AI-related job tasks (Acemoglu and Restrepo, 2018a; Autor et al., 2024).

⁴Most recently, the AI labs have started to release descriptive statistics on their internal usage data (Chatterji et al., 2025; Handa et al., 2025; Tomlinson et al., 2025). A complementary strength of our survey data is that we can capture the full range of AI chatbot products, link responses to external data sets (e.g., administrative labor market records), and sample both users and non-users from a well-defined population frame.

Finally, the null results across our 11 exposed occupations stand in contrast to some evidence from online freelance platforms, showing sharp declines in demand for tasks like writing and translation after ChatGPT’s release (Teutloff et al., 2025). Yet while some freelance tasks are highly substitutable with chatbots, our survey shows that most work—even in exposed occupations—is not, and that workers often reallocate saved time to other tasks. A key advantage of our administrative data is that it captures workers’ total earnings and hours, regardless of such reallocations.

1 Data and Institutional Setting

Denmark offers an ideal setting for examining the labor market impacts of Generative AI.

First, Danish workers have been at the forefront of Generative AI adoption, with take-up rates comparable to those in the United States (Bick, Blandin and Deming, 2025; Humlum and Vestergaard, 2025; RISJ, 2024).

Second, Denmark’s labor market is highly flexible, with low hiring and firing costs—similar to those of the U.S.—which allow firms and workers to adjust employment in response to technological change (Botero et al., 2004; Dahl, Le Maire and Munch, 2013). Appendix C.1 details wage-setting in our occupations, showing that most workers engage in annual negotiations with their employers, providing regular opportunities to adjust earnings and hours in response to AI chatbot adoption.

Third, Denmark has exceptional infrastructure for tracking the adoption of new technologies. In particular, every Dane has a digital mailbox that Statistics Denmark can use to distribute survey invitations. We use this infrastructure to conduct two large-scale, representative surveys on AI chatbot adoption.

Finally, our partnership with Statistics Denmark allows us to link these surveys to administrative matched employer-employee data, providing a unique opportunity to analyze labor market effects such as changes in earnings, working hours, and job mobility.

Taken together, these factors make Denmark a prime setting for observing early labor market effects of Generative AI, with insights that may extend to other advanced

economies—including the U.S.—and an unparalleled data infrastructure to assess these impacts rigorously.

1.1 Survey Data

This paper builds on two large-scale surveys on AI chatbot adoption, conducted in November–December of 2023 and 2024. The first survey provided the dataset for Humlum and Vestergaard (2025), which documented adoption patterns of ChatGPT, the dominant AI chatbot. This paper extends that dataset in two key ways. First, we link survey responses to administrative labor market data after the introduction of ChatGPT, enabling us to assess impacts on earnings, hours worked, and job mobility. Second, we introduce a second survey round in 2024 that *(i)* broadens the scope to include all AI chatbots, including custom versions, *(ii)* provides extensive data on employer-led adoption initiatives, and *(iii)* offers deeper insights into workers’ actual usage and perceived benefits of these tools. Appendix J outlines the 2024 survey, which is the primary focus of this paper.

1.1.1 Occupations

Our surveys focus on 11 occupations that are highly exposed to AI chatbots: *accountants, customer support specialists, financial advisors, HR professionals, IT support specialists, journalists, legal professionals, marketing professionals, office clerks, software developers, and teachers*. These occupations were selected based on three criteria: *(i)* they have at least one O*NET job task where AI chatbots can save time, as measured by the “Direct Exposure (E1)” metric from Eloundou et al. (2024); *(ii)* they are captured by a well-defined set of ISCO codes; and *(iii)* they contain a sufficient number of workers for statistical analysis. Humlum and Vestergaard (2025) details the selection and empirical measurement of these occupations. Together, these 11 occupations comprise the segments of the labor market where the impacts of AI chatbots are likely to emerge first and most strongly. Our analysis focuses on this set.